CHAPTER 10

NON-PARAMETRIC TESTS

CHAPTER ESSENCE AT A GLANCE:
Master distribution-free inference: advantages over parametric tests, One-Sample and Paired Sign Test, Wilcoxon Signed-Rank Test, and Wald-Wolfowitz Run Test for randomness.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Parametric vs Non-Parametric Tests:** Parametric tests assume specific population distributions (normality) and parameters (mu, sigma); Non-parametric tests make no distributional assumptions and are distribution-free.

• **Sign Test for Median:** Replaces numerical observations with plus and minus signs based on whether they exceed a hypothesized median; evaluates sign counts using Binomial B(n, 0.5) distribution.

• **Wilcoxon Signed-Rank Test:** Incorporates both direction (sign) and magnitude (rank of absolute differences); ranks absolute deviations |d|, attaches original signs, and tests rank sum T.

• **Run Test for Randomness:** Tests whether a sequence of binary qualitative attributes occurred randomly; a run is a succession of identical symbols; extreme few or extreme many runs indicate non-randomness.

ESSENTIAL STATISTICAL FORMULAS & MEASURES

$$\text{Sign Test Binomial Model} \iff X \sim B(n, 0.5) \quad [\text{under } H_0 : \text{Median} = M_0]$$

Binary Directional Count Non-Parametric Model

$$T = \min(T^+, T^-) \quad \text{where} \quad T^+ = \sum R_i^+, \quad T^- = \sum R_i^-$$

Wilcoxon Signed-Rank Test Sum Statistic

$$E(R) = \frac{2n_1n_2}{n_1+n_2} + 1 \quad \text{and} \quad \text{Var}(R) = \frac{2n_1n_2(2n_1n_2 - n_1 - n_2)}{(n_1+n_2)^2(n_1+n_2-1)}$$

Run Test Expected Runs and Dispersion Under Randomness

$$Z = \frac{R - E(R)}{\sqrt{\text{Var}(R)}} \sim N(0,1) \quad [\text{Large Sample Randomness Test}]$$

Asymptotic Normal Approximation for Sequence Runs

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In Sign Tests, observations exactly equal to hypothesized median M0 result in zero difference (d = 0); these must be discarded and sample size n reduced accordingly.

• **Alert:** Non-parametric tests can be used on nominal and ordinal rank data where parametric tests like t or Z cannot be computed.

• **Alert:** While non-parametric tests require fewer assumptions, they possess lower statistical power than parametric tests when population normality actually holds.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between Parametric Tests and Non-Parametric Tests on the basis of: (a) Population distribution assumption, (b) Scale of measurement, (c) Statistical power. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **(a) Distribution Assumption:** Parametric tests strictly require parent populations to follow a known distribution (usually Normal); Non-parametric tests make no distributional assumptions (distribution-free). **[1 Mark]**

• **(b) Scale of Measurement:** Parametric tests require interval or ratio data; Non-parametric tests can be used on nominal categories or ordinal ranked data. **[1 Mark]**

• **(c) Statistical Power:** When normality holds, parametric tests have higher power (less likely to commit Type II error); non-parametric tests have slightly lower power. **[1 Mark]**

2. Explain the step-by-step procedure of conducting a Sign Test for a single sample median. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **Step 1 (Formulate Hypotheses):** Null Hypothesis H0: Population Median = M0 versus Alternative Hypothesis H1: Median != M0. **[1 Mark]**

• **Step 2 (Assign Signs):** Compute difference (xi - M0) for each observation. Assign a plus sign (+) if xi > M0, and a minus sign (-) if xi < M0. Discard observations where xi = M0 and adjust n. **[1 Mark]**

• **Step 3 (Count Test Statistic):** Let S be the number of plus signs. Under H0, S follows Binomial distribution B(n, 0.5). **[1 Mark]**

• **Step 4 (Decision Rule):** Using binomial tables, compute probability of observing S. If probability <= alpha, reject H0; for large n (n > 25), use normal approximation Z = (S - 0.5n) divided by (0.5 * sqrt(n)). **[1 Mark]**

3. What is a Run Test? How is it used to test the randomness of a sequence of binary events? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Definition of a Run:** A run is an unbroken succession of identical symbols bordered by different symbols or no symbols at all (e.g. sequence H H T T T H has 3 runs). **[1 Mark]**

• **Testing Randomness:** Let n1 be number of heads and n2 be number of tails. Under H0 of randomness, the total number of runs R has expected value E(R) = 2*n1*n2 divided by (n1+n2) + 1. **[1 Mark]**

• **Decision:** Too few runs indicate clustering or trend; too many runs indicate systematic periodic alternation. If R falls outside critical boundaries, reject H0 (sequence is non-random). **[1 Mark]**

4. State any two advantages of Non-Parametric tests over Parametric tests. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Advantage 1:** Applicable when sample size is very small and the nature of the population distribution is completely unknown or non-normal. **[1 Mark]**

• **Advantage 2:** Computations are extremely simple and rapid, and they can directly analyze qualitative or ranked ordinal data. **[1 Mark]**

5. Why is the Wilcoxon Signed-Rank Test more powerful than the ordinary Sign Test? [Score: 3 Marks]

Answer & Key Valuation Steps:

• **Limitation of Sign Test:** The sign test considers only the direction (plus or minus) of differences and completely discards their numerical magnitudes. **[1.5 Marks]**

• **Wilcoxon Enhancement:** The Wilcoxon signed-rank test utilizes both the direction and the relative magnitude of differences by ranking absolute deviations |d|, thereby extracting more sample information and achieving higher statistical power. **[1.5 Marks]**

6. How are tied ranks handled in rank-based non-parametric tests? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **Handling Tied Ranks:** When two or more observations have identical values, each tied observation is assigned the average (mean) of the ranks they would have occupied if they were distinct. **[2 Marks]**